

considered for cooling of surfaces exposed to high-temperature flows [3]. Numerous experiments have been carried out since then, with both technological improvements on how wall transpiration is achieved, and the ambition of reducing the gap between idealized test cases in canonical flows and more complex scenarios with practical applications. The surface of the porous walls employed in early studies [4–6] had a significant influence on the boundary layer, limiting the applicability of this type of control to the cases for which a very strong transpiration intensity was possible. A significant advancement in this field was the development of construction procedures such as that employed for the microblowing technique (MBT) by Hwang [7], which enables the creation of relatively smooth porous surfaces. Subsequently, numerous studies reported the possibility of drag reduction with relatively low blowing intensities [8]. However, a more generalized application of this technology is yet to be achieved, a fact attributed, on the one hand, to implementation difficulties of engineering nature, and, on the other hand, to a lack of knowledge about how this kind of control would perform for complex turbulent flows [9].

These circumstances motivated the most recent experimental campaigns. Kornilov [10] carried out experiments on a Clark-Z airfoil at $Re_c = 840\,000$ (where Re_c is the Reynolds number based on incoming flow velocity U_∞ , chord length c , and fluid viscosity ν , defined as $Re_c = U_\infty c / \nu$) and various angles of attack, using a combination of suction at the leading edge and blowing on the pressure side of the airfoil. Subsequently, Eto *et al.* [11] considered uniform blowing on the suction side of a Clark-Y airfoil at angles of attack from 0° to 6° and $Re_c = 1\,500\,000$, using both active and passive blowing. The results of these two studies confirmed the relevance of test cases as close as possible to realistic scenarios because of the nonobvious effects that this control has on quantities such as total lift and drag. In particular, in both cases, only marginal reduction or even increase of the total drag were achieved. Kornilov *et al.* [12] employed a combination of uniform blowing on the pressure side and uniform suction on the suction side of the symmetric airfoil NACA0012 at a Reynolds number $Re_c = 700\,000$ and various angles of attack. In the same study, Reynolds-averaged Navier-Stokes (RANS) simulations were carried out, showing a qualitative agreement with the experiment and, in a later study, an estimate of energy cost for the same setup was provided [13].

High-fidelity numerical simulations have been also employed to study the effects of blowing and suction, but, due to their high computational cost, it has only recently become possible to study external flows at moderately high Reynolds numbers. The first direct numerical simulations (DNS) aimed at evaluating control strategies, including uniform blowing and suction, were carried out using turbulent channel flow as test case [14,15]. Note, however, that the most promising application of this type of control is in external flows. Furthermore, in these studies blowing and suction were also used as actuator mechanisms in feedback control, which uses information from sensors to modify the flow interactively, while this study is focused on predetermined control, i.e., with a fixed value of transpiration intensity.

The first DNS of a fully developed turbulent boundary layer (TBL) over a flat plate with uniform blowing and suction, which can be considered a model for a MBT transpiring surface, was performed by Kametani and Fukagata [16] but it was still limited to a very low Reynolds number $Re_\theta \approx 750$. Later on, Kametani *et al.* [17] performed a similar study with highly resolved large-eddy simulations (LES) extending the Reynolds-number range up to $Re_\theta = 2500$, also employing the so-called Fukagata-Iwamoto-Kasagi (FIK) decomposition [18] and spectral analysis to describe in detail how this control modifies the turbulent flow. At the same time, Stroh *et al.* [19], using opposition control as an example, discussed how control effects vary in turbulent channels and boundary layers. They reported that, even if a similar drag reduction is achieved, the contributions to the skin friction from the FIK decomposition are affected differently in both flows, suggesting that the interactions between control and turbulence may be qualitatively different for internal and external flows. Other studies explored topics such as differences and similarities of uniform blowing and opposition control with similar skin-friction reduction [20], how to better take into account the combined effects of blowing on skin friction, turbulent fluctuations, and boundary-layer thickness [21], and how an asymptotic state could be achieved for a developing turbulent boundary layer under suction [22,23].

term is related to TKE production, and the third term is related to the boundary-layer development:

$$c_{f1}^{\text{RD}} = \frac{2}{U_e^3} \int_0^{\delta_{99}} \nu \left(\frac{\partial U_t}{\partial y_n} \right)^2 dy_n, \quad c_{f2}^{\text{RD}} = \frac{2}{U_e^3} \int_0^{\delta_{99}} -\overline{uv} \frac{\partial U}{\partial y} dy, \quad c_{f3}^{\text{RD}} = \frac{2}{U_e^3} \int_0^{\delta_{99}} (U - U_e) \frac{\partial \tau}{\partial y} dy. \quad (\text{B1})$$

This last term can be also expressed as the sum of three contributions [43], denoted by $c_{f3}^{\text{RD}} = c_{f31}^{\text{RD}} + c_{f32}^{\text{RD}} + c_{f33}^{\text{RD}}$:

$$c_{f31}^{\text{RD}} = \frac{2}{U_e^3} \int_0^{\delta_{99}} (U - U_e) \left(U \frac{\partial U}{\partial x} + \nu \frac{\partial U}{\partial y} \right) dy, \quad c_{f32}^{\text{RD}} = \frac{2}{U_e^3} \int_0^{\delta_{99}} -(U - U_e) \left(\nu \frac{\partial^2 U}{\partial x^2} - \frac{\partial \overline{uu}}{\partial x} \right) dy, \\ c_{f33}^{\text{RD}} = \frac{2}{U_e^3} \int_0^{\delta_{99}} -(U - U_e) \left(\frac{1}{\rho} \frac{\partial p}{\partial x} \right) dy. \quad (\text{B2})$$

The RD contributions cannot be directly written in terms of the FIK identity, but it still is possible to verify whether the two decompositions lead to qualitatively different results by regrouping contributions that derive from the same terms of the governing equations. In particular, c_f^T and c_{f2}^{RD} are both related to the Reynolds shear stress, $c_f^{D1} + c_f^{D3}$ and c_{f31}^{RD} are both related to mean convection, $c_f^{D2} + c_f^{D4}$ and c_{f32}^{RD} are both related to streamwise derivatives, and c_f^P and c_{f33}^{RD} are both related to the pressure gradient. The relative proportion of these terms with respect to the total c_f is illustrated in Fig. 17 for the suction side of the reference cases at both $\text{Re}_c = 200\,000$ and $\text{Re}_c = 400\,000$. Although the corresponding contributions do not have the same numerical values, they exhibit a remarkably similar behavior. We observed that the same results also hold for the cases with control (not shown here), showing that our conclusions do not depend on the specific skin-friction decomposition chosen for the analysis.

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